Polymorphisms of SARS-CoV-2 in Washington State

Ishan Bansal[1]

[1]Department of Genome Sciences, University of Washington

Research Mentors: Dr. Kelley Harris[1] and Dr. Pengyao Jiang[1]

UNIVERSITY of WASHINGTON

Introduction

The global pandemic of SARS-CoV-2 has seen the virus evolving significantly as it has spread. This study presents a method and analysis of single nucleotide polymorphisms (SNP) across genomes primarily from Washington State. The SARS-CoV-2 genome consists of several domains, including a leading sequence, RNA replication proteins, non-structural proteins, and structural proteins. Another unique feature of note in coronaviruses is the prescience of replication proofreading mechanisms. This is important as the mutation spectrum differs from that of usual RNA viruses with a slower mutation rate. This study's findings can aid in the predictive efforts of the evolution of the virus and the development of a vaccine.

Methods

1. In this study, a total of 109 genomes were selected from the NCBI GenBank, comprising of three main geographic regions, China (with a specific emphasis upon those from Wuhan), Italy and Washington. The sequences from China and Italy are relatively few compared to that of Washington and merely serve to provide context and enhance the analysis of the single nucleotide polymorphisms found.
2. The sequences were then aligned (using NUCMER and MUSCLE) and SNPs were extracted.
3. To annotate the extracted SNP data and conduct more advanced analysis, a variety of tools were employed including SnpEff.
4. In addition, the use of Next Strain was crucial for plotting and generating the phylogenetic tree and MEGA for the substitution tests.

Polymorphisms In SARS-CoV-2

- Looking at the polymorphisms in SARS-CoV-2, the mutation spectrum found is shown in Figure 1, with a heavy C>T/U bias.
- In Figure 2, several SNPs that are dominant in the strains were analyzed (more than 50% as compared to reference).
  - This is the 4 base co-mutation, which has also been documented in Europe.
  - Also present, though to a lesser extent is the 5 pair co-mutation, a more unique spectra.
- Finally, in Figure 3, several proteins showed a high conservation ratio while others showed a high mutation ratio.

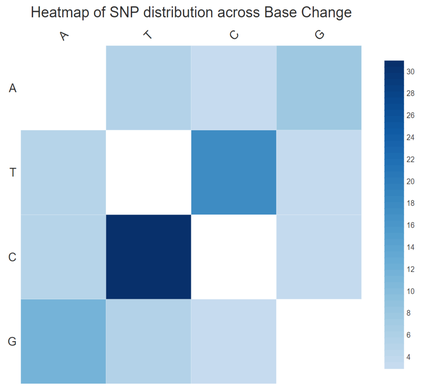

Figure 1: Heatmap of SNP distribution across 109 samples. The transition to transversion ratio found was 2.81

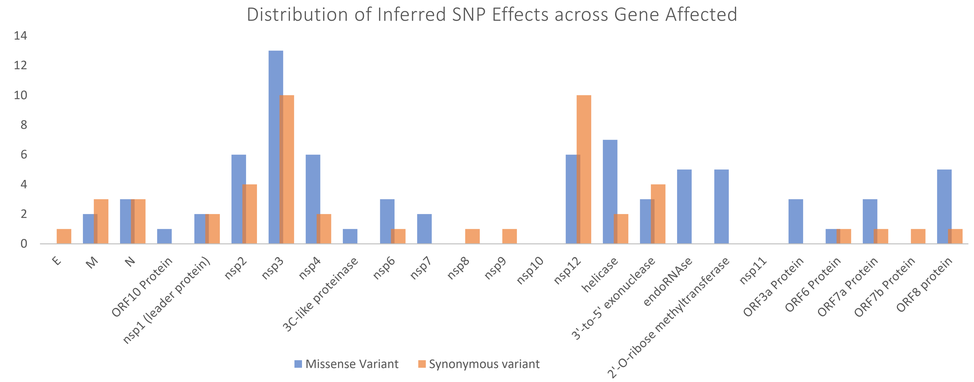

Figure 3: Inferred effects of the identified SNPs, classified by location and gene affected (does not account for downstream/upstream variants)

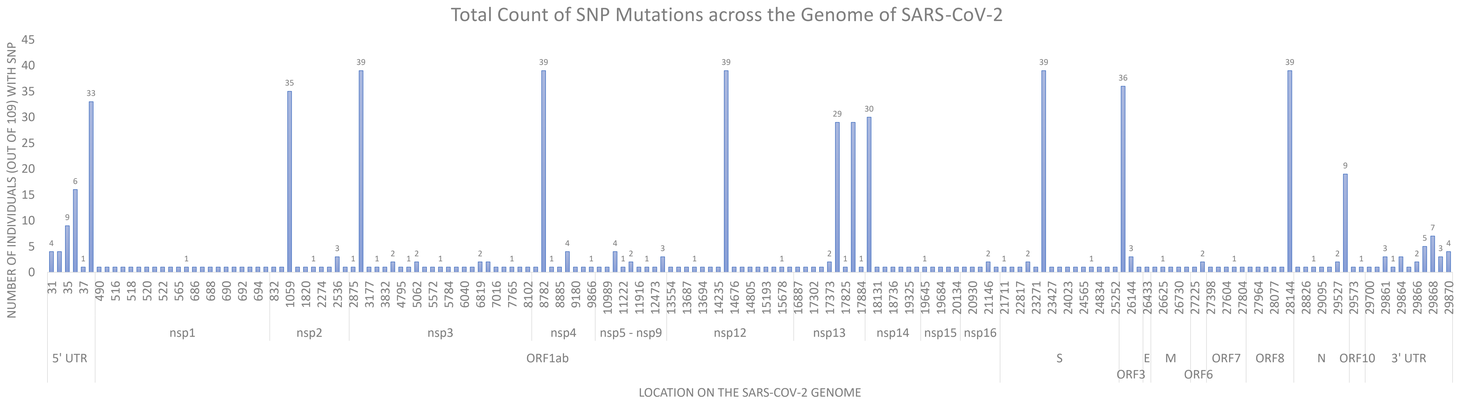

Figure 2: Frequency of SNPs by locus on the genome, categorized into genes across the 109 genomes sampled. Of interest: 4 base co-mutation: 241C>T, 3037C>T, 14408C>T and 23403A>G and a 5 base co-mutation: 8782C>T, 17747C>T, 17858A>G, 18060C>T, and 28144C>T

Phylogenetic Analysis

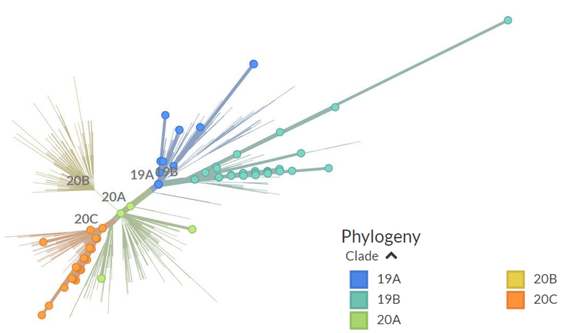

Figure 4: Phylogenetic tree of the 109 Samples using Nextstrain's Nextclade database. From here it is crucial to note that the Washington samples are predominantly in the three clades: 19A, 19B and 20C. 19A and 19B are direct descendants from the origin in Wuhan, 20A originated in the European cases, and 20B/20C are genetically unique clades.

The 4 base co-mutation found was also documented in European strains, and further analysis was done.

The phylogenetic analysis (nextclade) yielded the following results:

- The Washington samples exhibiting the 4 co-mutation are not of the European clade, making evolution more likely than IBD
- The 5 base co-mutation samples corresponded to the 20C clade, that may have arisen in the early Washington cases

Key Findings

- A 4 base co-mutation found in Europe was also discovered in Washington
- A new genetically independent clade, now prominent in the pandemic, arose in the early Washington cases study
- Certain proteins (nsp12, E, M, etc.) are being highly conserved and others (nsp4, helicase ORF8) are mutating

Acknowledgements

The author would like to acknowledge the University of Washington Genome Sciences, specifically Harris Lab, the global researchers who shared sequence data on the NCBI database, and the essential workers across the globe.

References

- Bedford, T., Greninger, A. L., Roychoudhury, P., & Starita, L. M. (2020). Cryptic transmission of SARS-CoV-2 in Washington State. *Medrxiv*. doi:https://doi.org/10.1101/2020.04.02.20051417
- Changchuan, Y. (2020). Genotyping coronavirus SARS-CoV-2: methods and implications. *Genomics*.
- Cingolani P, Platts A, Wang le L, Coon M, Nguyen T, Wang L, Land SJ, Lu X, Ruden DM. Fly (Austin). "A program for annotating and predicting the effects of single nucleotide polymorphisms, SnpEff"
- Hadfield *et al.*, Nextstrain: real-time tracking of pathogen evolution, *Bioinformatics* (2018)
- Marçais G, Delcher AL, Phillippy AM, Coston R, Salzberg SL, et al. (2018) MUMmer4: A fast and versatile genome alignment system. PLOS Computational Biology 14(1): e1005944.
- Y. Chen, Q. L. (2020). Emerging coronaviruses: genome structure, replication, and pathogenesis. *J. Med. Virol.*, 418-423.